Probabilistic and Temporal Machine Learning Approaches for 5G Handover Prediction and Analysis

Pattern Recognition and Machine Learning Project

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Presentation roadmap

1. Motivation and Problem Formulation
2. Dataset and Preprocessing
3. Temporal Targets and Models
4. Results
5. Conclusion

Problem Statement: 5G Handover Prediction

Problem Definition

Predict and analyse handover behaviour of mobile devices in a 5G network using historical network measurements, radio signal characteristics, and mobility-related information.

Given:

- Radio measurements such as RSRP(Reference Signal Received Power), RSRQ(Reference Signal Received Quality), and SINR(Signal-to-Interference-plus-Noise Ratio).
- Device mobility and geographical information.
- Current serving-cell information.
- Historical handover activity.

Our Prediction Objectives:

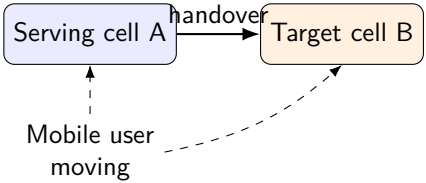
1. **Handover Classification:** Predict whether a sample belongs to the migration-positive class.
2. **Destination Cell Prediction:** Predict the serving cell associated with a network observation.

Main Research Objective

Develop machine learning models to understand and predict handover activity, particularly the number of future handovers within a given time horizon.

Why predict 5G handovers?

- A mobile device moves through the coverage regions of multiple base stations.
- A handover transfers an active connection from the serving cell to another cell.
- Poorly timed decisions can contribute to handover failure, interruption, ping-pong effects, or unnecessary signalling.
- Predictive mobility management can use mobility, radio, and network measurements to anticipate upcoming changes.



Central idea

Move beyond a single binary decision: estimate handover occurrence, destination cell, historical handover intensity, and future handover counts.

Six interconnected project objectives

1. **Handover warning:** classify whether the migration label is positive.
2. **Destination prediction:** predict the destination serving-cell class.
3. **Cumulative count:** quantify serving-cell transitions experienced so far.
4. **Historical counts:** count transitions in the preceding 30 s and 60 s.
5. **Handover rate:** express the previous 60 s count as events per minute.
6. **Future counts:** predict the number of transitions in the next 30 s and 60 s.

Important distinction: the binary label is derived from `migration > 0`; temporal events are derived from changes in `servingCell`. These are related but not assumed identical.

Mathematical Task Definitions

1. Handover Indicator

$$H_v(t_i) = \mathbb{1}\{C_v(t_i) \neq C_v(t_{i-1})\}$$

Indicates whether vehicle v changes its serving cell at time t_i. Returns 1 if a handover occurs, otherwise 0.

2. Cumulative and Historical Handovers

$$N_{\text{cum},v}(t) = \sum_{\tau \leq t} H_v(\tau)$$

Total number of handovers experienced by the vehicle up to time t.

$$N_{\text{past},v}(t, T) = \sum_{\tau \in (t-T, t]} H_v(\tau)$$

Number of handovers in the past T seconds (30 or 60 seconds).

3. Future Handover Count

$$Y_{v,T}(t) = \sum_{\tau \in (t, t+T]} H_v(\tau), \quad T \in \{30, 60\}$$

Number of handovers expected in the next 30 or 60 seconds. Used as the regression target.

4. Prediction Tasks

- **Binary classification:** Predict whether a vehicle will migrate (0 = No, 1 = Yes).
- **Destination classification:** Predict the destination cell of the vehicle.
- **Historical features:** Past 30-second count, past 60-second count, and cumulative handover count.
- **Count regression:** Predict future handover counts for 30 and 60 seconds.

Krakow dataset: four simulation runs

Run	Observations	Vehicles
900_1	259,918	775
1000_1	307,879	887
1200_1	373,688	1,045
1400_1	466,786	1,218
Combined	**1,408,271**	**3,925**

- Source used in the notebook: `FUMD-AI/krakow-handover-dataset`.
- Combined table: 1,408,271 rows and 28 columns after selecting features and adding run/group identifiers.
- Each run's `veh_id` is local to that run; `group_id = run_id + veh_id` prevents collisions.
- Nine destination/serving-cell labels are represented (cells 1–9).

Feature Groups and Their Abbreviations

Mobility Features

- Position: x, y (Coordinates)
- Speed and Direction
- Position along trajectory (pos)
- Signal Indicator

Network Performance

- RLC Downlink Delay (Radio Link Control)
- RLC PDU Delay (Protocol Data Unit)
- RLC Throughput (Radio Link Control)

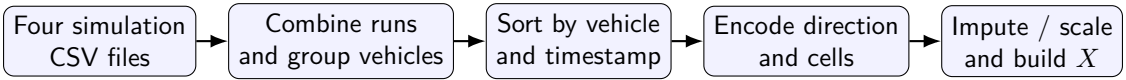
Radio Features

- Downlink CQI (Channel Quality Indicator)
- Distance
- Measured DL SINR (Downlink Signal-to-Interference-plus-Noise Ratio)
- Measured UL SINR, Received DL SINR

Cell Context and Labels

- Current Serving Cell
- Previous Serving-Cell History
- Migration Indicator
- Destination Cell

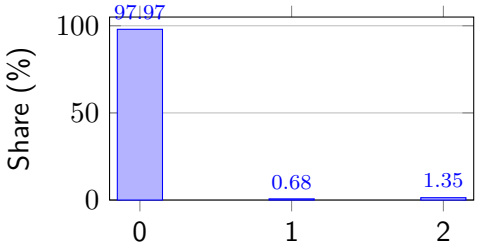
Preprocessing and feature construction



- Direction is circular: $\theta_{\sin} = \sin(\theta\pi/180)$ and $\theta_{\cos} = \cos(\theta\pi/180)$.
- This avoids treating 359° and 1° as far apart.
- 15 numerical features + 42 one-hot categorical features = **57 features**.
- Notebook check: zero missing values and zero infinite values in the constructed base feature matrix.

Migration-label imbalance

Migration class	Count	Share
0	1,379,632	97.966%
1	9,574	0.680%
2	19,065	1.354%



Implication

A majority-class predictor obtains 97.97% accuracy Accuracy alone is therefore misleading.

Temporal event construction

- Within each vehicle group, sort observations chronologically.
- A transition event is recorded when the serving cell differs from the previous observation.
- Use timestamps and event-time arrays to count events in past/future windows.
- `searchsorted` enables efficient boundary lookup in sorted event timestamps.
- Rows without a complete future 60-second observation window are excluded from future-count training.

Leakage control

All temporal features are calculated within a vehicle group. Group-aware splitting keeps a vehicle's observations from being distributed across train, validation, and test sets.

PRML models used

Task	Models	Role
Binary handover warning	Gaussian NB, LDA, QDA, Logistic Regression, Bayesian Logistic Regression, GMM	Estimate probability of positive migration label
Destination cell	Gaussian NB, LDA, QDA, Logistic Regression, GMM	Predict one of nine destination cells
Future counts (30/60 s)	Poisson Regression, Histogram Gradient Boosting	Predict non-negative count targets

- NB: Bayes rule with conditional feature independence.
- LDA/QDA: Gaussian class-conditional densities; shared vs class-specific covariance.
- Logistic regression: discriminative posterior $P(y = 1 \mid x) = \sigma(w^\top x + b)$.
- GMM: latent mixture components estimated using expectation-maximization.

GMM: probabilistic mixture modelling

A Gaussian mixture models the feature density as

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k).$$

- E-step: compute component responsibilities $r_{ik} = P(z_i = k \mid x_i)$.
- M-step: update mixture weights, means, and covariance parameters.
- Binary handover: estimate smoothed component handover rates and combine them with responsibilities.
- Destination: estimate component-to-cell probabilities, then compute

$$P(d = c \mid x_i) \approx \sum_{k} r_{ik} P(d = c \mid z = k).$$

Notebook configurations

Binary GMM: 8 components, diagonal covariance, `reg_covar` $= 10^{-4}$, maximum 100 iterations; fitted on a random subset capped at 50,000 rows. Destination GMM: 15 selected

Experimental protocol

- Group-aware train/validation/test partitioning is used to reduce vehicle-level leakage.
- Validation data is used for classification threshold selection and model selection.
- Test data is reserved for final performance reporting.
- Binary classification: probability thresholds are selected on validation data; thresholded test predictions are then evaluated.
- Destination prediction: multiclass probabilities are evaluated using accuracy, balanced accuracy, macro-F1 and log loss.
- Temporal regression: only samples with complete future windows are retained.

Temporal split sizes from notebook

Train: 763,571 Validation: 175,661 Test: 237,784

Temporal model input dimension: 60 features (57 base + 3 historical handover features).

Binary handover prediction: test results

Model	Precision	Recall	F2	PR-AUC
Bayesian Logistic Reg.	0.1541	0.7043	0.4109	**0.2194**
LDA	0.1322	0.6524	0.3652	0.2173
QDA	0.1639	0.7411	**0.4348**	0.2158
Logistic Regression	0.1400	**0.8370**	0.4193	0.2013
Gaussian Naive Bayes	0.0718	0.4923	0.2267	0.0748
GMM	0.0545	0.4125	0.1783	0.0720

- Bayesian Logistic Regression has the highest test PR-AUC (0.2194).
- QDA has the highest F2 (0.4348); Logistic Regression has the highest recall (0.8370).
- The selected model depends on the operational objective: ranking quality, recall, or recall-weighted F-score.

Binary classification: interpreting the trade-off

Why PR-AUC?

With only about 2.03% positive migration labels, the precision-recall curve is more informative about positive-class detection than accuracy alone.

Why F2?

$$F_2 = 5\frac{PR}{4P + R}$$

F2 weights recall more heavily than precision, appropriate when missed handovers are costly.

Baseline

Majority-class baseline:

$$\text{Accuracy} = 0.9593,$$

Takeaway

The baseline's high accuracy is not evidence of useful handover detection.

Destination serving-cell prediction

Model	Accuracy	Balanced Acc.	Macro-F1
Logistic Regression	**0.8965**	**0.9196**	**0.9024**
QDA	0.8544	0.9003	0.8668
LDA	0.8411	0.8469	0.8395
Gaussian Naive Bayes	0.5269	0.6763	0.5453
GMM	0.5627	0.5406	0.4919

- Logistic Regression is the strongest tested destination classifier by macro-F1 and log loss (0.2721).
- Destination GMM selected 15 components using validation log loss; test accuracy = 0.5627 and macro-F1 = 0.4919.
- Destination GMM top-3 test accuracy = 0.9726, indicating that the true cell is often among the three highest-ranked candidates.

Future handover-count prediction

Horizon	Model	MAE	RMSE	R^2	Poisson Dev.
30 s	Poisson Regression	0.7493	1.3152	0.1749	1.4792
30 s	Hist. Gradient Boosting	**0.4261**	**0.9659**	**0.5549**	**0.6402**
60 s	Poisson Regression	1.2433	1.9267	0.1935	2.1476
60 s	Hist. Gradient Boosting	**0.8012**	**1.4757**	**0.5269**	**1.1086**

- Histogram Gradient Boosting outperforms Poisson Regression at both horizons.
- Error increases for the 60-second horizon, as expected for a longer forecast window.
- Nonlinear interactions between mobility, radio conditions, and historical handovers can benefit from tree boosting.

Handover rate and temporal interpretation

- Historical handover rate is the count in the previous 60 seconds, interpreted as events per minute.
- Future predicted counts are converted to a rate by

$$\widehat{\lambda}_{\text{per min}} = \frac{\widehat{Y}_T}{T} \times 60.$$

- Mean predicted future rate for 60-second horizon:
  - Poisson Regression: 1.1352 handovers/min.
  - Histogram Gradient Boosting: 1.1332 handovers/min.
- Predicted rate distributions can help identify vehicles likely to experience frequent cell transitions.

Interpretation

A count prediction is not a handover decision by itself. It provides an estimate of near-future mobility-related network transition intensity.

Key findings

1. The project formulates handover analysis as linked classification and temporal count-regression tasks.
2. The dataset contains 1.408 million observations from 3,925 vehicles across four simulation runs.
3. Binary migration prediction is challenging because the positive class is rare; PR-AUC and recall-sensitive metrics are essential.
4. Logistic Regression gives the best destination-cell macro-F1 (0.9024).
5. Histogram Gradient Boosting gives the best future count predictions for both 30 s and 60 s horizons.
6. Historical handover counts and cumulative transitions provide temporal context for future predictions.

Limitations and future work

Current limitations

- Simulation dataset; external-network generalization is not established.
- Severe binary class imbalance.
- Count targets depend on timestamp semantics and event construction.
- GMM destination model trails discriminative classifiers on macro-F1.

Future directions

- Sequence models and trajectory-aware learning.
- Probability calibration and cost-sensitive thresholds.
- Spatial context and network topology.
- Real-time inference and online adaptation.
- Evaluation on additional environments and unseen mobility patterns.

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- FUMD-AI, "Krakow handover dataset," GitHub repository, `https://github.com/FUMD-AI/krakow-handover-dataset`.

Thank you

Questions?